DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE489
Name: MD Ashikul Islam
ID: 24241376
Email: md.ashikul.islam1@g.bracu.ac.bd

Project Title: Zello A Flutter-Based E-Commerce Mobile Application

Project Features:
Zello is a full-stack mobile e-commerce application built with Flutter and powered by the Firebase platform. It supports two distinct user roles Customer and Admin each with a dedicated dashboard and feature set. The following features have been implemented in this project:

A. User / Customer Features

a) **User Registration & Login:** Users can create an account with email and password or sign in with Google. Firebase Authentication manages sessions so users remain logged in across app restarts.

b) **Home Dashboard with Categories:** After login, users see a home screen displaying product categories created by the admin (e.g., Electronics, Clothing, Grocery) as scrollable cards, along with a promotional banner carousel and new arrivals section.

c) **Browse & Search Products:** Users can browse the full product catalog displayed as a grid or list, and search products by name. Results update in real time as the user types.

d) **Filter & Sort:** A filter bottom sheet allows users to narrow results by category, price range, minimum star rating, and in-stock availability. Sort options include price (low/high), newest, and top-rated.

e) **Product Detail Page:** Tapping a product opens a full detail view with a swipeable image gallery, name, price, description, available variants (size/color), stock status, and average rating. Stock count is a live Firestore stream, updating without manual refresh.

f) **Stock Status & Out-of-Stock Handling:** Products with zero stock display a prominent "Out of Stock" badge and the Add to Cart button is disabled. When stock is between 1 and 5, an amber "Only N left!" warning is shown. Users can tap "Notify Me" to receive a push notification when the product is restocked.

g) **Cart Management:** Users can add products to cart with selected variant and quantity. The cart is stored in Firestore and persists across devices. The app bar shows a live badge count. Items can be removed with a swipe gesture.

h) **Wishlist:** A heart icon on product cards and detail pages allows users to save products to a wishlist stored in Firestore. Saved items can be moved to cart in a single tap.

i) **Checkout:** The checkout screen displays the delivery address (editable), selected payment method, item summary, and order total including shipping. Payment is

processed via SSLCommerz (to be integrated). An order document is created in Firestore only upon confirmed payment success.

j) **Order Tracking:** Users can view all their past and active orders. A visual step indicator shows the current status (Pending → Confirmed → Shipped → Delivered) and updates live via a Firestore stream listener whenever the admin changes the status.

k) **Product Ratings & Reviews:** After an order is delivered, users are prompted to leave a 1–5 star rating with an optional text comment and photo. Only verified buyers can submit reviews, preventing fake feedback. Average ratings are recalculated automatically via a Firebase Cloud Function.

l) **User Profile:** Users can view and edit their name, upload a profile photo (stored in Firebase Storage), manage saved delivery addresses, and configure notification preferences.

m) **Dark Mode:** The app supports a system-aware dark mode using Flutter's ThemeData. It automatically follows the device's light or dark system preference, with a manual override option available in the profile settings screen.

B. Admin Features

n) **Admin Dashboard:** After login with an admin account, the admin is routed to a dedicated panel showing summary statistics: total registered users, total products, orders placed today, and revenue today.

o) **Category Management:** Admins can create, edit, and delete product categories. Each category has a name and a cover image uploaded to Firebase Storage. Categories appear immediately on the user's home screen upon creation.

p) **Add & Edit Products:** Admins can add new products by entering name, description, price, stock quantity, category (selected from a Firestore dropdown), product images (multi-upload to Firebase Storage), and optional variants such as size or color. Existing products can be fully edited at any time.

q) **Update Price & Stock:** A quick-edit button on each product card in the admin list allows rapid price and stock updates without opening the full product edit form. Out-of-stock products are flagged with a red badge for easy identification.

r) **Order Management & Status Updates:** Admins can view all orders, filter by status, and update each order through the stages: Pending → Confirmed → Shipped → Delivered or Cancelled. Status changes trigger a Firebase Cloud Function that sends a push notification to the customer.

s) **User List & Management:** The admin panel includes a User List showing all registered customers with name, email, join date, and order count. Admins can disable an account (blocking login) or fully remove a user from the system.

C. Technical Implementation

t) **Flutter UI:** A single Dart codebase runs natively on both Android and iOS. Navigation is handled by GoRouter with role-based redirect guards. Reusable widgets (product card, status badge, shimmer loader) are organized in a shared components folder.

u) **Firebase Authentication:** Handles all identity management including email/password login, Google Sign-In, password reset, and persistent sessions. A role field in each user's Firestore document determines whether they are routed to the user dashboard or admin panel.

v) **Cloud Firestore Database:** A NoSQL real-time database used to store all application data. Key collections include: users, products, categories, orders, carts, reviews, coupons, and notifications. Security rules restrict read/write access based on user role.

w) **Firebase Storage:** Used to store all binary files including product images, category images, promotional banners, user profile photos, and review photos. Download URLs are saved in corresponding Firestore documents.

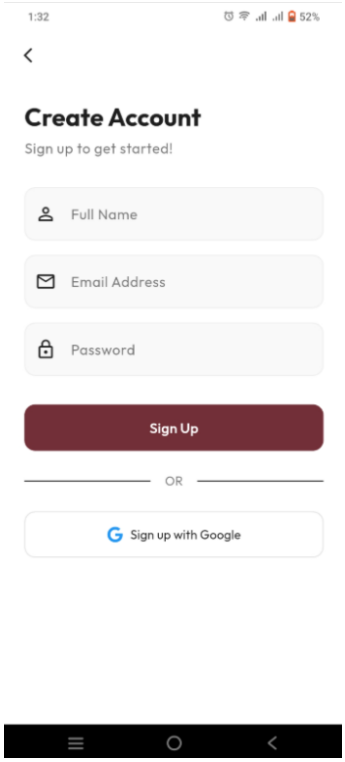
x) **State Management (Riverpod):** Riverpod is used for global state management across the app. Each feature (auth, cart, products, filters) has its own AsyncNotifier or StateNotifier, ensuring compile-time safety and clean separation of concerns.

y) **Firebase Cloud Functions & FCM:** Server-side Cloud Functions handle automated tasks such as recalculating product ratings after a review, sending push notifications via Firebase Cloud Messaging (FCM) on order status changes and restock events, and user deletion using the Admin SDK.

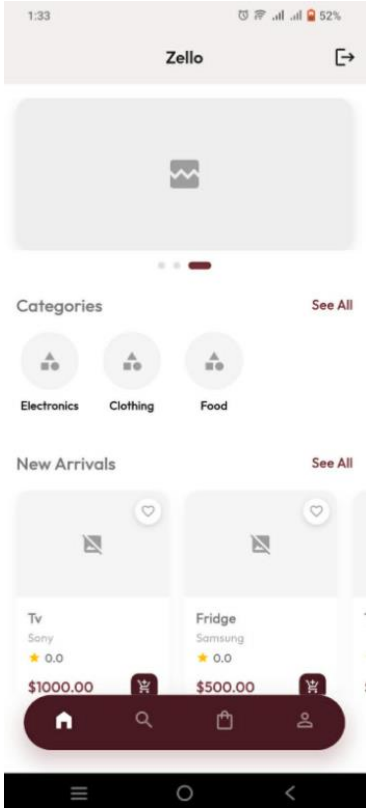
***Screenshots*:**

The following screenshots demonstrate the core screens of the Zello application:

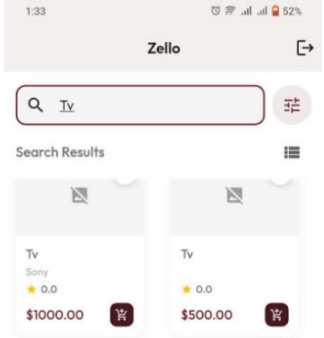
z) Login & Registration Screen

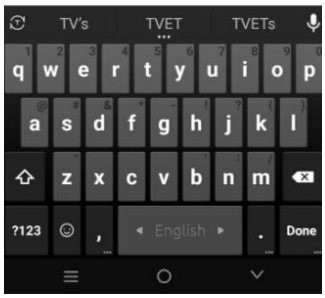


aa) Home Dashboard with Categories & Banner

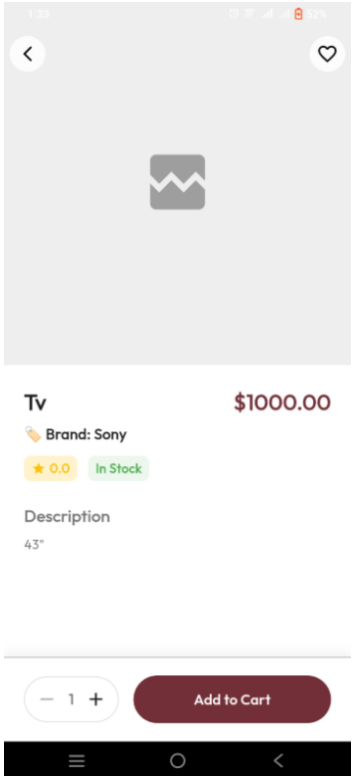


bb) Product Catalog with Filter & Search

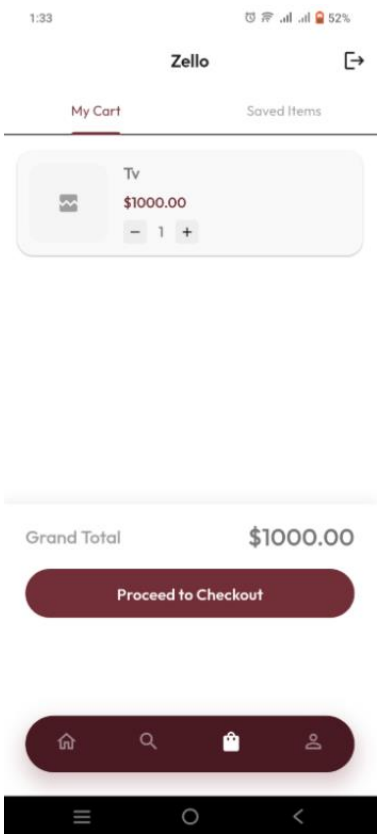


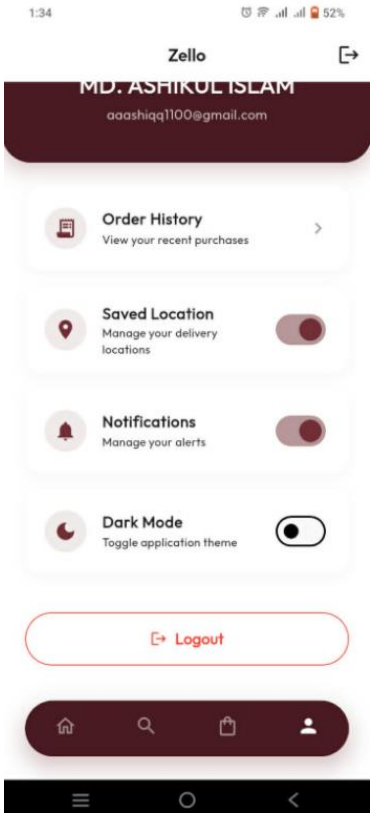


cc) Product Detail Page (In Stock & Out of Stock states)

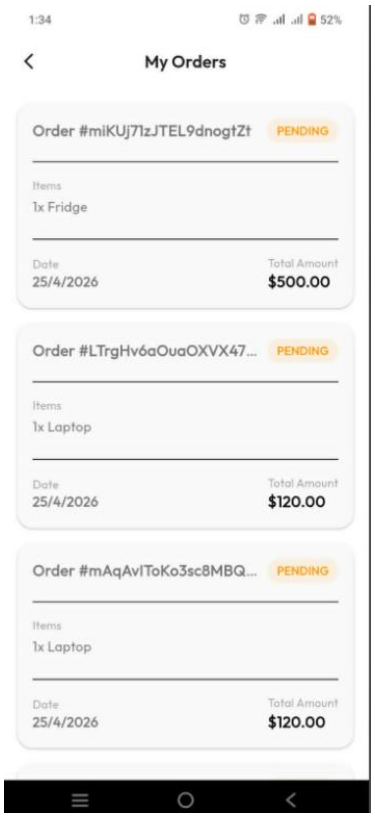


dd) Cart & Checkout Screen

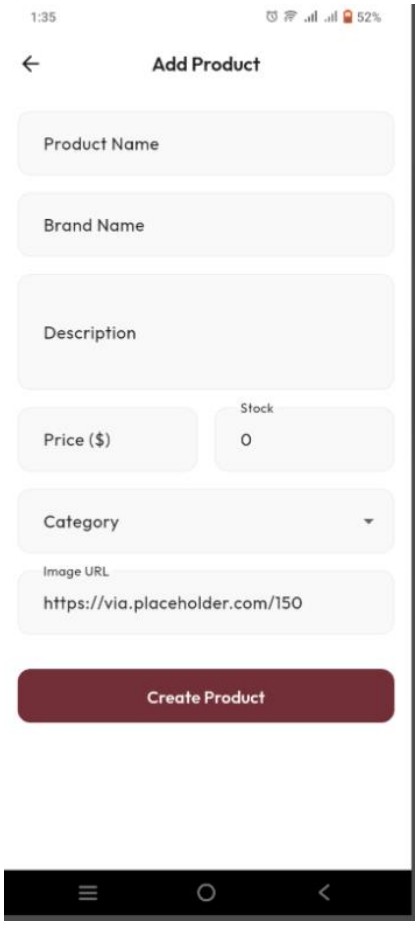




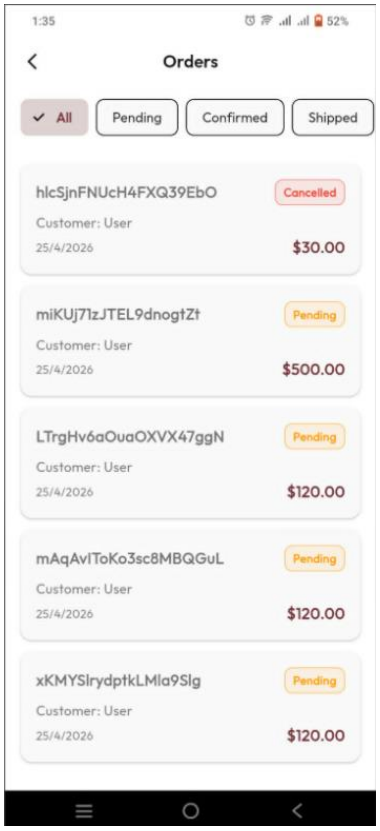
ee) Order Tracking with Status Timeline

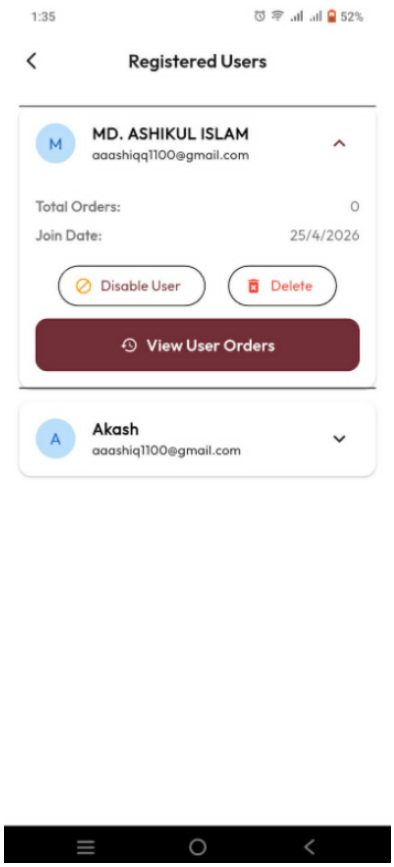


ff) Admin Panel — Product Management



gg) Admin Panel — Order Status Update & User List





Online Resources used:

a) *Official Documentation & References:*

- *Flutter Official Documentation — https://docs.flutter.dev*
- *Firebase Documentation (Auth, Firestore, Storage, Cloud Functions, FCM) — https://firebase.google.com/docs*
- *Riverpod State Management Documentation — https://riverpod.dev/docs*
- *GoRouter Navigation Package — https://pub.dev/packages/go_router*
- *SSLCommerz Payment Gateway (Future Integration) — https://developer.sslcommerz.com/doc/v4*
- *Flutter Pub.dev Package Repository — https://pub.dev*

b) *Video Tutorials:*

- *Flutter & Firebase Full E-Commerce App — https://www.youtube.com/@johannes*
- *milke*
- *Riverpod for Beginners — https://www.youtube.com/@RivaanRanawat*

c) *Community & Code References:*

- *Stack Overflow — Flutter and Firebase tagged questions — https://stackoverflow.com/questions/tagged/flutter*
- *GitHub — flutter/flutter repository & community packages — https://github.com/flutter/flutter*

Future Enhancements:

The following enhancements are planned for future development phases to further improve the system:

a) **SSLCommerz Payment Integration:** Full integration of the SSLCommerz payment gateway to support bKash, Nagad, Rocket, and card payments, enabling a complete end-to-end purchase flow within the Bangladeshi payment ecosystem.

b) **Promo Codes & Discount System:** Admin-created coupon codes (percentage or flat discount) with expiry dates and usage limits, redeemable at checkout.

c) **Multi-Language Support (Bangla & English):** Using Flutter's intl package and ARB localization files to support both Bangla and English, improving accessibility for a wider Bangladeshi user base.

d) **Product Sharing via Dynamic Links:** Users can share a product page link that opens the app directly on that product, or falls back to a web preview if the app is not installed.

e) **Firebase Analytics Integration:** Tracking screen views, add-to-cart events, search queries, checkout funnel drop-offs, and purchase completions to enable data-driven improvements to the app.

f) **Advanced Admin Reporting:** Graphical sales reports, revenue charts by category, top-selling products list, and low-stock alerts within the admin panel.